

Machine Learning

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Lecture 04: K Nearest Neighbor

Sources

- **Machine Learning for Intelligent Systems**, Kilian Weinberger, Cornell University, Lecture 2, https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote02_kNN.html
- **Nearest Neighbor Methods**, Victor Lavrenko, Assistant Professor at the University of Edinburgh, https://www.youtube.com/playlist?list=PLBv09BD7ez_48heon5Az-TsyoXVYOJtDZ
- **Wiki K-Nearest Neighbors**: https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm
- **Effects of Distance Measure Choice on KNN Classifier Performance - A Review**, V. B. Surya Prasath et al., <https://arxiv.org/pdf/1708.04321.pdf>
- **A Comparative Analysis of Similarity Measures to find Coherent Documents**, Mausumi Goswami et al. <http://www.ijamtes.org/gallery/101.%20nov%20ijmte%20-%20as.pdf>
- **A Comparison Study on Similarity and Dissimilarity Measures in Clustering Continuous Data**, Ali Seyed Shirkhorshidi et al., <https://journals.plos.org/plosone/article/file?id=10.1371/journal.pone.0144059&type=printable>

KNN – The good, the bad and the ugly

- KNN is a simple algorithm but is highly effective for solving various real life classification problems. Especially when the datasets are large and continuously growing.
- **Challenges:**
 1. How to find the optimum value of K?
 2. How to find the right distance function?
- **Problems:**
 1. High computational time cost for each prediction.
 2. High memory requirement as we need to keep all training samples.
 3. The curse of dimensionality.

The Curse of Dimensionality

(https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote02_kNN.html)

- **KNN Assumption**

- Points near one another tend to have the same label

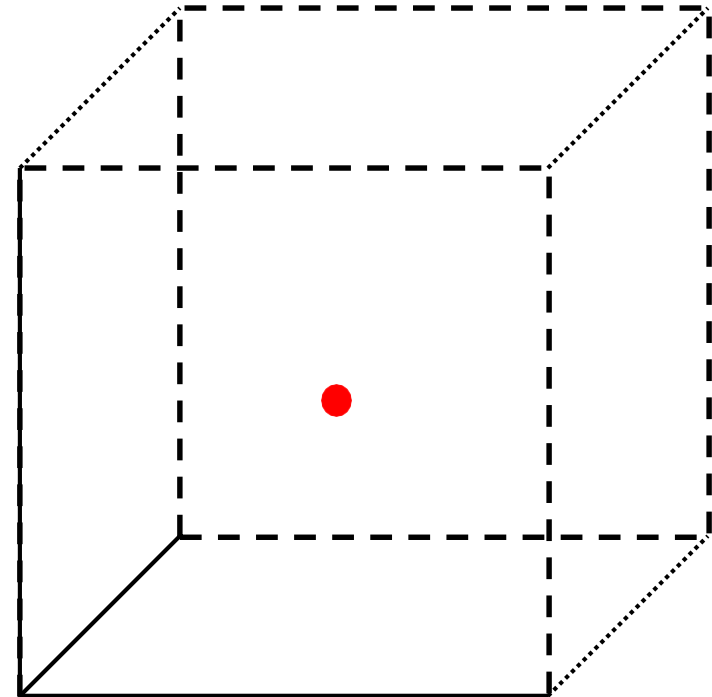
- **Problem**

- In high dimensions, points drawn from a uniform distribution, are never near one another
 - So, no neighbors near you!

Demonstration

(https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote02_kNN.html)

- Let's draw points at random (sampled from a uniform distribution) from the unit cube
- How much space the k nearest neighbors of the red test point will occupy?



Demonstration

(https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote02_kNN.html)

- Imagine the unit cube $[0,1]^d$
- All training data is sampled uniformly within this cube

$$\forall i, x_i \in [0,1]^d$$

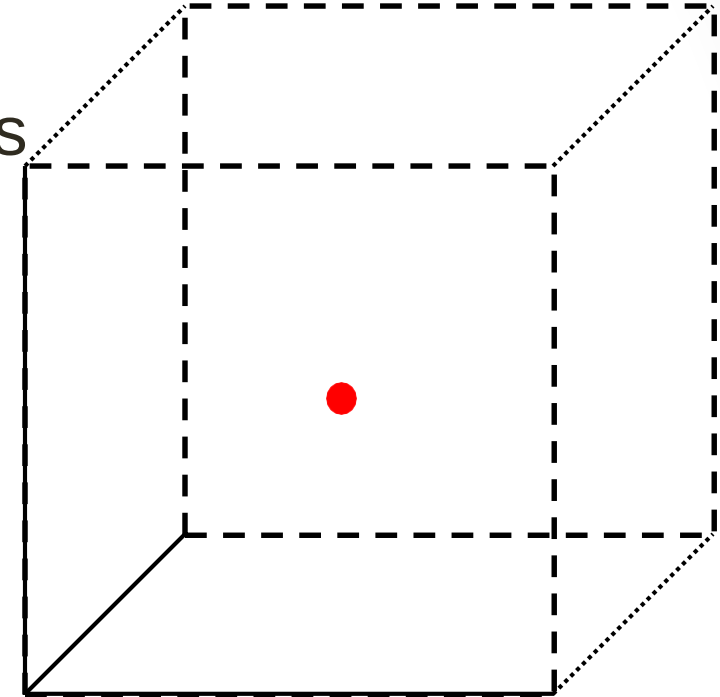
- Now, let l be the edge length of the smallest hyper-cube that contains the k nearest neighbor of the test point

$$l^d \approx \frac{k}{n}$$

- As the points are drawn uniformly

$$l = \left(\frac{k}{n}\right)^{\frac{1}{d}}$$

- If $k = 10$ and $n = 1000$. How big is l ?



d	l
2	0.1
10	0.63
100	0.955
1000	0.9954

Demonstration

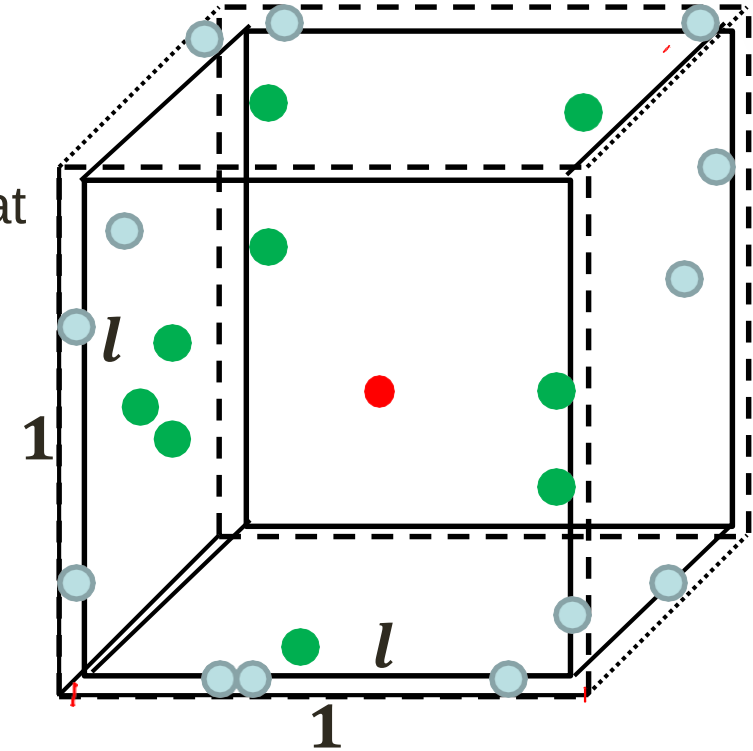
(https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote02_kNN.html)

- Imagine the unit cube $[0,1]^d$
- All training data is sampled uniformly within this cube
- Now, let l be the edge length of the smallest hyper-cube that contains the k nearest neighbor of the test point

$$\forall i, x_i \in [0,1]^d$$

- As the points are drawn uniformly
- $$l^d \approx \frac{k}{n}$$
- $$l = \left(\frac{k}{n}\right)^{\frac{1}{d}}$$

- If $k = 10$ and $n = 1000$. How big is l ?
- So the test point has no neighbors near it!
- Most “nearest” neighbors (green) are close to the edges otherwise we would have drawn a smaller cube around them
- All “non-nearest” neighbors are squished between the two cubes, very near to the “nearest neighbors” So why should the test point share its label with these points? The whole premise fails!



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So how does KNN work at all?

(https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote02_kNN.html)

- Many practical problems (SPAM vs. Not SPAM with bag-of-words, face recognition) etc. have *high dimensionality* (SPAM vs. Not SPAM has $|V|$, face recognition may have [number of pixels]
- e.g. 18M) but *low intrinsic dimensionality*
 - May be a small subset of $|V|$ helps with the SPAM vs. Not SPAM classification
 - May be the actual face recognition features are much less e.g. humans recognize face based on m/f, color, hair color, spectacles, shape of the nose etc.
- So the true or effective dimensionality might be much lower making KNN still feasible.

Thank You